Structural rigidity in the Erdős–Graham two-set permutation problem

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Abstract

A sequence $a_1, a_2, \ldots$ *contains a monotone 3-AP* if some three terms, taken in order of position, form an increasing or a decreasing arithmetic progression. Davis, Entringer, Graham and Simmons (1977) proved that every permutation of $\mathbb{Z}^+$ contains an increasing — in particular monotone — 3-AP, constructed a partition of $\mathbb{Z}^+$ into three sets each admitting a permutation with no monotone 3-AP, and asked whether two sets suffice; Erdős and Graham (1979, p. 338) repeated this as "a very annoying question" (problem #197 of the Erdős problem collection). We prove several structural results constraining any solution. (i) A permutable set admits no infinite doubling orbit $u_{k+1} = 2u_k - f_k$ with f_k drawn from a fixed finite subset. (ii) In any 3-AP-free arrangement, every newly placed value must be balanced: the counts of already-placed values in $[1, v)$ and in $(v, 2v)$ control each other up to the codensity of the set in $(v, 2v)$ and in $(0, v)$ respectively. (iii) No 3-AP-free arrangement of $\{k, 2k, 3k, 5k\} \subseteq S$ places all ratio-≥ 2 pairs in increasing order. (iv) For the canonical dyadic partition, no solution can play any cofinite family of dyadic blocks as contiguous runs: a halving self-reduction plus finitely many machine-checked base cases shows the associated "zone" systems are infeasible, and a finish-time descent completes the argument. (v) In contrast, all natural finite fragments of the problem are satisfiable, and we give an exact characterization of solvability in terms of bounded-delay towers of finite arrangements, which pinpoints the problem's content in the growth of a single computable function. (vi) For the dyadic partition we prove an unconditional reduction to a single scale-uniform statement: an "order gadget" $\mathrm{OG}(M)$ on the interval $(M, 2M]$ — no monotone in-block 3-AP, plus fifteen guard-precedence axioms induced by the values 15 and 16 — whose infeasibility at infinitely many dyadic scales $M = 2^{2t-1}$ implies that S_A admits no valid permutation. (vii) We then prove, by hand, that infeasibility: for every $M \equiv 0 \pmod 8$ with $M \geq 16$, already the three-axiom core $\mathrm{C3} \subset \mathrm{OG}(M)$ is inconsistent with AP-freeness. The proof runs on a small toolkit of ladder lemmas (zigzag propagation, phase dichotomy, a transfer lock, and a mirror-flood induction) and exhibits the mod-8 arithmetic as a residue condition on two flood centers; every step is machine-verified at 100 scales up to $M = 1024$ and independently cross-validated. Combining (vi) and (vii): S_A **admits no permutation free of monotone 3-APs** — the canonical dyadic partition does not resolve the Erdős–Graham problem. We also prove a clean absorption theorem for van der Corput orderings of odd residue

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*With extensive machine assistance from Claude (Anthropic) and from the SAT solvers CaDiCaL and OR-Tools CP-SAT; SAT certificates, code, and a one-command reproduction kit are available in the accompanying repository (see Data Availability). All proofs have been verified by hand unless explicitly marked as machine-checked.

classes which explains the fractal structure observed in all satisfying fragments. Problem #197 itself remains open: our theorem eliminates one candidate partition, and decides nothing about other two-set partitions.

1 Introduction

Call a linear arrangement of a set of integers *admissible* if no three of its terms, read in order of position, form an increasing or a decreasing arithmetic progression (a *monotone 3-AP*). Every finite set of integers admits an admissible arrangement: place the odd values before the even values and recurse, using the fact that the first and third term of a 3-AP share a parity (stated for intervals in [1], which credits earlier notes of this observation; for arbitrary finite sets in [2]). For infinite sets the situation changes completely. Davis, Entringer, Graham and Simmons [1] proved that every permutation of $\mathbb{Z}^+$ (of order type ω) contains an increasing 3-AP — in particular a monotone one — so $\mathbb{Z}^+$ itself admits no admissible permutation. They also constructed a partition of $\mathbb{Z}^+$ into *three* sets, each admitting an admissible permutation, and asked (Concluding Remarks 3, p. 89) whether two sets suffice. Erdős and Graham repeated the question in [2] (p. 338), calling it "a very annoying question" (see also their monograph [3]); it is problem #197 of the Erdős problem collection [10], where it is stated for $\mathbb{N}$ and carries a Lean formalisation [11].

The modern literature around the problem concerns densities and longer progressions. LeSaulnier and Vijay [5] exhibited a set admitting an admissible permutation with upper density 1/2 and lower density 1/4; writing $\alpha_{\mathbb{N}}(3)$ (resp. $\beta_{\mathbb{N}}(3)$) for the supremum of upper (resp. lower) densities of such sets, this gives $\alpha_{\mathbb{N}}(3) \geq 1/2$ and $\beta_{\mathbb{N}}(3) \geq 1/4$, and they conjectured both bounds sharp. Geneson [9] recently disproved the upper-density conjecture, proving $\alpha_{\mathbb{N}}(3) \geq 2/3$. His witness is strikingly relevant here: it is a union of ratio-4 doubling blocks $[L \cdot 4^j + m,\, 2L \cdot 4^j]$ — an S_{A}-like octave pattern *with the bottom sliver of each block removed* and only finitely many octaves per stage. Our main theorem shows the full dyadic blocks fail, and the mechanism of failure (the attack of the values 15 and 16 on every block's bottom sliver, Section 11 onward) lives precisely in the region Geneson's construction excises: the two results are consistent, and the mechanisms align strikingly: the attacks proving failure of the full blocks are concentrated in the same bottom regions removed in Geneson's construction. (We do not claim that removing precisely those slivers is necessary or sufficient; his construction also uses stage separation and rapid scale growth.) Permutations of $\mathbb{Z}$ and of subsets of $\mathbb{Z}^+$ avoiding longer monotone progressions were constructed by Geneson [6] and Adenwalla [7]. Hirose and Saito [8] — continuing a line begun by Ardal, Brown and Jungić [4] on AP-avoiding ("chaotic") orderings of $\mathbb{Q}$ and $\mathbb{R}$ — characterised the *order types* of total orders on $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ admitting no monotone 3-AP; from their standpoint the difficulty of #197 is that order type ω is forced, where avoidance is hardest.

The natural first candidate for a two-set partition — implicit already in the parity heuristic of [1] — is the *dyadic partition*: let

$$S_{\mathrm{A}} \;=\; \bigcup_{k \text{ even}} \left(2^{k-1},\, 2^k\right], \qquad S_B \;=\; \mathbb{Z}^+ \setminus S_{\mathrm{A}},$$

so that each team is a union of doubling blocks and the completion $2y - x$ of a within-block pair $x < y$, whenever it escapes the block upward, lands in the adjacent block — the other team's territory. Our main result eliminates this candidate.

Main Theorem (Theorem 35). *S_A admits no permutation of order type ω free of monotone 3-term arithmetic progressions. Consequently the dyadic partition does not witness a positive answer to problem #197.*

The proof has an unusual architecture, which we believe is of independent interest. First, an exact *chunk reduction* (Theorem 19) translates the existence of an admissible permutation into the existence of a stage function with finite fibres satisfying two checkable conditions; in these coordinates the theorem of [1] (in its monotone form) has a four-line proof, and self-similarity of S_A under $v \mapsto 4v$ becomes an induction engine. Second, a sequence of machine-assisted eliminations (Sections 11–12) shrinks the search space of candidate stage functions until a single finite obstruction emerges: an *order gadget* $\mathrm{OG}(M)$ on the interval $(M, 2M]$, fifteen precedence axioms induced by the two values 15 and 16, whose infeasibility at infinitely many dyadic scales implies the main theorem unconditionally (Theorem 30). We stress that Theorem 30 is proved directly on a hypothetical permutation and is logically independent of the chunk calculus: the chunk coordinates served as the discovery instrument, and give an alternative route to the same reduction (Remark 31), but the proof chain of the main theorem is just Theorem 30 plus Theorem 36. Third — and this is the mathematical heart — we prove by hand that already a three-axiom core of $\mathrm{OG}(M)$ is inconsistent with admissibility for every $M \equiv 0 \pmod 8$ (Theorem 36), via a toolkit of ladder lemmas whose single appeal to the residue of M occurs at two "flood centres" near $3M/2$. (The basic zigzag alternation on AP-ladders that seeds the toolkit is classical — it appears inside [1] in Folkman's argument, as Lemma 2.5 of [8], and as Lemmas 2.1–2.2 of [9]; what is new here is its boundary-quantitative use on bounded ladders, the transfer lock, and the flood induction.) Computationally, the mod-8 condition is sharp at every tested scale: the same three axioms are satisfiable at every tested scale with $M \not\equiv 0 \pmod 8$ (a parametric satisfying construction for the complementary residues is not proved here, and the main theorem does not require one), matching the machine's phase diagram exactly.

A word on methodology. The elimination phase and the discovery of the gadget, the three-axiom core, and the residue phenomenon were carried out with large-scale SAT-solver experimentation (CaDiCaL, OR-Tools), with every negative claim cross-validated by independent encodings and solvers. The final proof chain, however, does not rest on any solver verdict: the reductions and the C3 core theorem are human-readable arguments, and each of their finitely many schema instances has been machine-audited by three independently written checkers, including mutation testing of the checkers themselves. A complete reproduction kit (scripts, pinned environments, and DRAT certificates for the finite base cases) accompanies the paper.

Problem #197 itself remains open. Beyond eliminating the leading candidate, the methods here — the chunk calculus, the attack/guard analysis of small values against doubling blocks, and the residue-locked ladder toolkit — apply to any two-set partition whose parts are unions of intervals, and we expect them to bear on the general question.

2 Definitions and conventions

Definition 1. A set $S \subseteq \mathbb{Z}^+$ is *permutable* if there is a bijection $\pi\colon \mathbb{N} \to S$ such that no $i < j < k$ have $\pi(i), \pi(j), \pi(k)$ forming an arithmetic progression with $\pi(i) < \pi(j) < \pi(k)$ or $\pi(i) > \pi(j) > \pi(k)$.

The completion of an ordered pair: if x is placed before y, the value $z = 2y - x$ (which continues the progression through x, y in the direction of y) must not be placed after y; equivalently every such $z \in S$ placed later yields a monotone 3-AP. We use throughout the *dyadic blocks* $B_k = (2^{k-1}, 2^k]$ and the canonical two-set candidate $S_{\mathrm{A}} = \bigcup_{k \text{ even}} B_k$, $S_B = \mathbb{Z}^+ \setminus S_{\mathrm{A}}$.

Remark 2 (Ground set: $\mathbb{Z}^+$ versus $\mathbb{N} \ni 0$; the Lean formalization)**.** The original sources pose the problem for the positive integers: Davis, Entringer, Graham and Simmons ask whether a two-set partition of $\mathbb{Z}^+$ is possible, and Erdős–Graham (1979, p. 338) repeat the question, again for $\mathbb{Z}^+$. The Lean formalization in the DeepMind *formal-conjectures* repository (`FormalConjectures/ErdosProblems/197.lean`) instead partitions Lean's $\mathbb{N} = \{0, 1, 2, \dots\}$, which *includes* 0, and asks for bijections $f \colon \mathbb{N} \simeq A$, $g \colon \mathbb{N} \simeq B$ avoiding monotone 3-term APs — the bijection-from-$\mathbb{N}$ requirement being exactly our order-type-ω convention. The two formulations are not verbatim equivalent: a positive answer over $\mathbb{N}$ yields one over $\mathbb{Z}^+$ (delete the value 0; deletion preserves 3-AP-freeness), but conversely adjoining 0 to one part adds genuine constraints — the progressions $(0, d, 2d)$ with $d, 2d$ in that part — and we are not aware of a general argument that a permutable set stays permutable after adjoining 0. Our results are insensitive to the nuance: Theorem 35 shows S_{A} itself is not permutable, and since deletion preserves validity, $S_{\mathrm{A}} \cup \{0\}$ is not permutable either; hence the dyadic partition fails in both formulations, whichever part receives 0. (In the Lean encoding the values live in $\mathbb{N}$, so the common difference is a natural number and never negative; `HasMonotoneAP` requires strictly increasing indices whose value list equals $(a, a+d, a+2d)$ or its reverse, the reversed disjunct being the decreasing case. The predicate permits $d = 0$, but injectivity of the permutation rules that out, so the notion of "monotone 3-AP" agrees with ours.)

3 The orbit obstruction

Lemma 3 (Orbit obstruction)**.** *Let S be permutable and $F \subseteq S$ finite. There is no infinite sequence $u_0 < u_1 < \cdots$ in S with $u_{k+1} = 2u_k - f_k$, $f_k \in F$.*

Proof. Let q be the largest position of an element of F in the permutation. Since the u_k are distinct, at most q of them occupy positions $\le q$; beyond the largest such index, each pair (f_k, u_k) is increasingly placed, so its completion $u_{k+1} \in S$ must be placed before u_k, giving an infinite strictly decreasing sequence of positions. □

Taking $S = \mathbb{Z}^+$, $F = \{1, 2\}$ recovers the theorem of Davis–Entringer–Graham–Simmons (more precisely its monotone form; [1] in fact proves that an *increasing* 3-AP always occurs).

4 The balance law

Lemma 4 (Balanced placement)**.** *Let S be permutable via π, and consider the moment a value v is placed. Let L (resp. H) be the number of already-placed values in $[1, v)$ (resp. $(v, 2v)$). Then*

$$L - H \;\le\; |S^c \cap (v, 2v)| \qquad \textit{and} \qquad H - L \;\le\; |S^c \cap (0, v)| + O(1).$$

(The two directions are bounded by the codensities of S in different ranges; no symmetric single-range bound holds in general.[1]*)*

Proof. Each placed $x < v$ forms an increasingly placed pair (x, v) with completion $2v - x \in (v, 2v)$, distinct for distinct x; each completion lying in S must already be placed. This gives $H \geq L - |S^c \cap (v, 2v)|$. In the other direction, each placed $u \in (v, 2v)$ forms a decreasingly placed pair (u, v) with completion $2v - u \in (0, v)$, distinct for distinct u, and each such completion lying in S must already be placed, whence $L \geq H - |S^c \cap (0, v)| - O(1)$. □

Corollary 5 (Records)**.** *When a left-to-right maximum v is placed, at most $|S^c \cap (v, 2v)| + O(1)$ smaller values are already placed.*

5 Two absorption lemmas

Lemma 6 (Ratio-ascent obstruction)**.** *Let $S \supseteq \{k, 2k, 3k, 5k\}$. No 3-AP-free arrangement of S places w before u for every pair $u \geq 2w$ (elements of S).*

Proof. The hypothesis forces $k \prec 2k$, $k \prec 3k$, $2k \prec 5k$. The progression $(k, 2k, 3k)$ with $k \prec 2k$, $k \prec 3k$ forces $3k \prec 2k$; then $3k \prec 2k \prec 5k$, and $(k, 3k, 5k)$ is an increasing 3-AP. □

Theorem 7 (van der Corput absorption)**.** *Order the odd residues c modulo 2^k by the bit-reversal (van der Corput) rank of $(c-1)/2 \bmod 2^{k-1}$. Then for any two classes a ordered before b, the class $2b - a \bmod 2^k$ is ordered strictly before b.*

Proof. Write $\alpha = (a-1)/2$, $\beta = (b-1)/2$; the completion class has parameter $\zeta = 2\beta - \alpha \bmod 2^{k-1}$. If the lowest bits of α, β agree, then so does ζ's, and dividing by 2 preserves the form $\zeta' = 2\beta' - \alpha'$. At the first bit where they differ, the assumption $\mathrm{rev}(\alpha) < \mathrm{rev}(\beta)$ forces α's bit to be 0 and β's to be 1; since $\zeta \equiv \alpha$ at that bit, ζ's bit is 0, so $\mathrm{rev}(\zeta) < \mathrm{rev}(\beta)$ strictly. □

6 Contiguous blocks cannot work

Definition 8. The *zone system* $Z(M)$: arrange the interval $(M, 2M]$ so that (a) no monotone 3-AP occurs, and (b) for every $y < z$ in $(M, 2M]$ with $2y - z \in (M/4, M/2]$, z precedes y.

Condition (b) is forced when the block $(M/4, M/2]$ is entirely placed before a contiguous run of $(M, 2M]$ within a team's sequence (its elements' completions $2y - x$ then must precede y).

Lemma 9 (Halving descent)**.** *If $Z(M)$ is satisfiable then so is $Z(\lfloor M/2 \rfloor)$.*

Proof. Restrict a witness to the even values of $(M, 2M]$ and halve; arithmetic progressions and the zone constraints scale exactly (the boundary cases are routine), and a superfluous top element may be deleted. □

Lemma 10 (Base cases; machine-checked)**.** *$Z(M)$ is unsatisfiable for $16 \leq M \leq 31$.*

[1] We thank Jesse Geneson for correcting an earlier symmetric misstatement of this lemma.

Theorem 11 (No contiguous-run solutions)**.** *In any 3-AP-free permutation of S_{A}, infinitely many blocks B_{2k} are not placed as contiguous runs.*

Proof. Lemmas 9 and 10 together make $Z(M)$ unsatisfiable for every $M \geq 16$ (halve repeatedly into $[16, 31]$). Suppose for contradiction that every block B_{2k} with $k \geq k_0 \geq 3$ is placed as a contiguous run. Contiguous runs are disjoint segments of the permutation, so for each $k > k_0$ one of the runs B_{2k-2}, B_{2k} entirely precedes the other. If B_{2k-2}'s run came first, then the whole block $(M/4, M/2]$, $M = 2^{2k-1}$, would be placed before the run of $(M, 2M]$, forcing condition (b) of the zone system as explained above — so the run of B_{2k} would witness $Z(M)$ with $M \geq 16$, which is unsatisfiable. Hence B_{2k}'s run precedes B_{2k-2}'s for every $k > k_0$, and the finish positions of the runs of $B_{2k_0}, B_{2k_0+2}, B_{2k_0+4}, \ldots$ form an infinite strictly decreasing sequence of positive integers — impossible. $\square$

7 Local realizability and the tower characterization

Definition 12 (Pure-complete systems)**.** For an S_{A}-block top X, the *pure-complete-X system* asks for an arrangement of $S_{\mathrm{A}} \cap [1, X]$ with no monotone 3-AP. (All completions of in-range pairs lie in $(X, 2X]$, an odd block, or inside the range; hence this is exactly the restriction of the infinite problem.)

Lemma 13 (Horizon collapse)**.** *Deleting any set of values from a valid arrangement preserves validity; hence auxiliary values above X never aid the pure-complete-X system.*

Theorem 14 (Tower characterization)**.** *S_{A} is permutable iff there exists $B\colon \mathbb{N} \to \mathbb{N}$ such that for every block top X the system [pure-complete-X and every v has at most $B(v)$ predecessors] is satisfiable.*

Proof. ($\Rightarrow$) restrict a permutation and take $B(v)$ its positions. ($\Leftarrow$) the restriction sets are finite and nested; Kőnig's lemma yields a consistent tower whose limit has each v preceded by at most $B(v)$ values, hence order type ω; all constraints are finitary and inherited. $\square$

Computationally: pure-complete-X is satisfiable for $X \leq 1024$ (machine certificates), and the completion-pressure function $g_X(L) = \min \max_{v \leq L} \operatorname{pos}(v)$ satisfies $g_{256}(64) = 153$ (optimal): eighty-seven percent of the block $(128, 256]$ must precede the completion of $[1, 64]$, with the residue concentrated in a single residue class modulo 8 — the reservoir structure that any construction must implement. Divergence of $g_X(L)$ in X for fixed L would prove non-permutability; the mechanism is blocked by the measured subset-tolerance (near-total), and g is expected to stabilize.

7.1 Self-similar witnesses, fragility, and pumping obstructions

Adding the exact scale-invariance $u \prec w \iff 4u \prec 4w$ to the pure-complete systems remains satisfiable at $X = 256$ and $X = 1024$ (independently audited witnesses); however the audited 1024-witness does not extend to a self-similar 4096-arrangement (machine-certified infeasibility), so the self-similar tower also has dead branches.

We also record a calibration fact: requiring, in addition, that the ± 1 neighbors (within the same block and team) of each forced completion also precede the pair's later element (“radius-1 robustness”) is infeasible in every setting we tested — including plain intervals $[1, n]$ from $n = 8$. Radius-1 robustness is thus unachievable generically in three-AP arrangement problems and cannot serve as a satisfiability criterion; in particular it does not discriminate between partitions, and pumping arguments must instead manage rounding drift on the thin family of far-pair constraints only. The radius-one experiments are *evidence* against placement rules in which position varies slowly with value; we neither formalize such a class of rules here nor prove a theorem about it, and we draw no further inference from the observation.

8 Partition rigidity

Lemma 15 (Ray piercing). *Let $\mathbb{Z}^+ = A \sqcup B$ with both parts permutable. Then for every $a \in A$ and every $d \geq 1$, the set $\{k \geq 0 : a + 2^k d \in B\}$ is infinite, and symmetrically with A and B exchanged.*

Proof. If $a + 2^k d \in A$ for all $k \geq k_0$, the values $u_j = a + 2^{k_0+j} d$ form an infinite orbit $u_{j+1} = 2u_j - a$ inside A with $F = \{a\}$, contradicting Lemma 3. □

Every affine doubling ray based in one team therefore meets the opposite team infinitely often (the precise content of the lemma — it does not assert literal alternation); partitions such as “even 2-adic valuation vs. odd” die immediately, and viable partitions are forced toward dyadic-interval alternation, motivating the canonical candidate S_{A}.

9 Suffix-stacked class deferral

Definition 16 (Deferral states). For a scale M (a power of two) let $D_M \subseteq (M/2, M]$ denote a single residue class modulo $m(M)$ (class coherence; the witnesses select $\{v \equiv 2 \bmod 2^{k/2}\}$). A *deferral state* at scale M is a doom-free arrangement of $S_{\mathrm{A}} \cap [1, 2M]$ of the suffix-stacked form $[\,\text{bulk} \mid D_{M/2} \mid D_M\,]$: the bulk contains everything except the two most recent deferred classes, which are appended in scale order.

Machine-verified facts (certificates in the repository): deferral states exist for $M = 32, 64, 128$; the suffix must be stacked in scale order (the reversed stacking contains a Lemma 6 configuration); no-extras variants are infeasible at every scale (lookahead into the next block is necessary); any single residue class serves as the deferred suffix, while generic sets of the same size do not.

Proposition 17 (Class-sink algebra). *Fix a modulus $m = 2^k$ and a target class r. The attacker map $\varphi_r(b) = 2b - r$ on $\mathbb{Z}/m\mathbb{Z}$ is conjugate to doubling (via $b \mapsto b - r$), so for m a power of two every orbit is eventually absorbed at the unique fixed point r: the functional graph is a tree onto r. Hence in a class-phase arrangement a single class can be scheduled last (all its attacker-demands point inward), whereas a fully placed zone imposes root demands at every class simultaneously, which cannot all be satisfied. For general m (odd part > 1) the conjugacy shows the cycle structure of φ_r is independent of r, and the tree picture fails; the observed special role of $2 \cdot \mathrm{odd}$ phases in the experiments is a range-truncation phenomenon, not a cycle-structure one.*[2]

[2] We thank Jesse Geneson for correcting the general-m form of this proposition; an earlier version asserted an r-dependent cycle structure.

10 The chunk reduction

Every permutation of S_{A} of order type ω decomposes into consecutive finite segments (*chunks*); conversely, a choice of finite fibers and internal orders assembles into such a permutation. The following exact reformulation is the coordinate system for the elimination phase (Sections 11–12); as noted in the introduction, the reduction of Section 13 is proved without it.

Definition 18. A *stage function* is any $s\colon S_{\mathrm{A}} \to \mathbb{N}$ with finite fibers. Given s and a linear order on each fiber, let T be the concatenation.

Theorem 19 (Chunk reduction; exact)**.** *T avoids monotone 3-term APs if and only if:*

(A) no AP triple (x, y, z), $z = 2y-x$, in S_{A} has strictly monotone stages ($s(x) < s(y) < s(z)$ or $s(z) < s(y) < s(x)$); and

(B) each fiber order contains the forced pairs

$$\begin{aligned} s(x) < s(y) = s(z):\ z \prec y, \qquad s(z) < s(y) = s(x):\ x \prec y, \\ s(x) = s(y) < s(z):\ y \prec x, \qquad s(y) = s(z) < s(x):\ y \prec z, \end{aligned}$$

and is non-monotone on every within-fiber triple. (Triples whose endpoint stages lie on one side of the middle's are automatically safe.)

Hence S_{A} is 3-permutable if and only if some stage function satisfies (A) and (B). The fiber problems are finite and mutually independent given s.

Proof. Along positions of T the stage is non-decreasing, and within a stage the fiber order gives the positions. Suppose T avoids monotone 3-APs. If (A) failed at (x, y, z) with $s(x) < s(y) < s(z)$, then x, y, z occur in T in that position order with $x < y < z$ — an increasing 3-AP (the case $s(z) < s(y) < s(x)$ gives a decreasing one). If a forced pair of (B) failed, say $s(x) < s(y) = s(z)$ with $y \prec z$ in their fiber, then x, y, z occur in position order — again an AP; the other three rows and the within-fiber non-monotonicity requirement are identical. Conversely, suppose (A) and (B) hold and let (x, y, z), $z = 2y-x$, be placed monotonically in T, say $\mathrm{pos}(x) < \mathrm{pos}(y) < \mathrm{pos}(z)$ (the decreasing case is symmetric). Then $s(x) \le s(y) \le s(z)$. Strict inequalities throughout contradict (A); $s(x) = s(y) < s(z)$ puts $x \prec y$ in one fiber, contradicting the row $s(x) = s(y) < s(z) \Rightarrow y \prec x$ of (B); $s(x) < s(y) = s(z)$ puts $y \prec z$, contradicting $z \prec y$; and $s(x) = s(y) = s(z)$ is a monotone within-fiber triple, excluded by (B). Triples whose endpoint stages lie on one side of the middle's stage can never be placed monotonically, since the middle's position is then extremal among the three; this is the safety clause. The full case table over all stage patterns and fiber orders has additionally been machine-checked exhaustively (`e96_reduction_check.py`, part P2). □

Lemma 20 (Normalization: running-max chunking)**.** *Every 3-AP-free permutation π of S_{A} (of order type ω) admits a valid scheme — a stage function with finite fibers satisfying (A)∧(B), whose concatenation is π itself — with $s(v) \ge \mathrm{block}(v)/2$ for every v.*

Proof. Set $s(v) = \max_{q \le \mathrm{pos}(v)} \mathrm{block}(\pi(q))/2$ (the running maximum of half-block-indices). Along positions s is non-decreasing, so its fibers are consecutive segments of π and the concatenation (with the fiber orders induced by π) is π; Theorem 19 then gives (A)$\wedge$(B). Since $\text{block}(v)/2$ enters the maximum at v's own position, $s(v) \geq \text{block}(v)/2$. Each fiber is finite: π is onto S_{A}, so a value of block index $> 2\sigma$ occurs at some finite position, after which the running maximum exceeds σ. □

The point of Lemma 20 is that non-negative *displacement* (next section) is a harmless normalization, not an automatic property: an arbitrary chunk decomposition can place a high-block value in an early chunk. Restrictions of normalized schemes stay normalized, which is what the divergence criterion below consumes.

Theorem 21 (DEGS via chunks). *$\mathbb{Z}^+$ itself is not 3-permutable. Proof: suppose s satisfies (A) with finite fibers and put $\sigma = s(1)$. For every y with $s(y) > \sigma$, the triple $(1, y, 2y-1)$ forces $s(2y-1) \leq s(y)$ by (A). The orbit y_0, $y_{i+1} = 2y_i - 1$ is strictly increasing; only finitely many of its members can have stage $\leq \sigma$ (those fibers are finite), after which its stages are non-increasing, hence eventually confined to one finite fiber — impossible for an infinite set.* □

This recovers the Davis–Entringer–Graham–Simmons theorem (in its monotone form; [1] proves the stronger increasing-3-AP statement) in the chunk coordinates and isolates what a permutable team must do: break every such reflected orbit out of the team. The dyadic team S_{A} breaks single-reflector orbits in two steps (completions land in odd blocks); the question is whether the multi-scale accounting below can be survived.

Lemma 22 (Same block). *In every AP triple of S_{A} the middle y and top z lie in the same even block. ($z < 2y \leq 2^{k+1}$ and block $k+1$ is odd.)*

Lemma 23 (Escape). *If $y \geq 3 \cdot 4^{s-1}$ (top quarter of block $2s$) and $x \leq 4^{s-1}$, then $2y - x > 4^s$: completions of (lower value, top-quarter value) pairs land in the odd block above — out of team.*

Lemma 24 (Bottom-half characterization). *Deferring $D_k \subseteq$ block k by uniform delay is (A)-clean if $D_k \subseteq (2^{k-1}, 3 \cdot 2^{k-2}]$: the witness midpoints $y = (x+z)/2$ of a bottom-half z against lower-block x satisfy $2^{k-2} < y \leq 2^{k-1}$, i.e. lie in the odd block below. Conversely a deferred top-half z has genuine in-team witnesses, forcing its witness midpoints (bottom-quarter values) to defer with it.*

11 The negative atlas

Theorem 25 (Block-granular death). *No stage function whose fibers are unions of whole even blocks satisfies (A)$\wedge$(B). Proof: the fatal core F_b (forced pairs into block b from block $b-2$ below, plus block-b triples) is UNSAT for $b = 6$ (machine, instant), and $v \mapsto 4v$ embeds F_b into F_{b+2}: the map preserves S_{A}, sends block j to block $j+2$, and preserves AP triples; UNSAT lifts to supersystems. Any block-granular scheme has a fiber boundary above block 4 and therefore contains some F_b.* □

Theorem 26 ($\times 4$ restriction; window death is permanent). *For the window class $\{s : s(v) - \text{block}(v)/2 \in [0, w]\}$: feasibility at horizon $4N$ implies feasibility at horizon N (restrict along $v \mapsto 4v$ and shift stages by one). Consequently the machine verdicts below are permanent* *(hold at all larger horizons), and an infinite solution in a window class would imply feasibility at every finite horizon.*

Machine atlas (CP-SAT and CDCL cross-checks; certificates in the repository): bottom-half relief of depth 1 and depth ≤ 2 infeasible at $N = 256$ (and 1024); **window-2 infeasible at** $N = 1024$ (CP-SAT, 746s; independently CDCL/unary-ladder, 1760s) — hence dead at every horizon; window-2 at $N = 256$ is feasible only via whole-block merges (minimum total displacement 41), whose skeleton is $C_s = 2^s \cdot [3 \cdot 2^{s-2}, 2^s]$, the top-quarter multiples of 2^s — precisely the set singled out by Lemma 23.

The surviving candidate shapes for a YES are stage functions with unbounded displacement (chunks mixing stragglers of unboundedly many blocks); the surviving NO program is to close the gap between Theorem 25/26 and general finite-fiber schemes via the per-block deferral calculus of Lemma 24. Both are finite-structure questions; the same lemmas hold verbatim for S_B (machine-verified mirror at 4096).

12 The displacement ladder

For a stage function s write $\delta(v) = s(v) - \text{block}(v)/2$ (*displacement*); a scheme is *normalized* if $\delta \geq 0$ everywhere, which by Lemma 20 is no loss of generality for schemes induced by genuine permutations. Define

$$L(m) \;=\; \min_{\text{normalized schemes at horizon } 4^m} \; \max_{v \leq 16} \delta(v),$$

the minimum over all normalized finite-fiber schemes satisfying (A)$\wedge$(B) at horizon 4^m (no bound on any other value's displacement).

Proposition 27 (Ladder rungs; machine-certified, solver-cross-checked). *$L(3) \leq 1$; $L(4) = 2$ (infeasibility of the cap $\delta \leq 1$ on $\{v \leq 16\}$ at $N = 256$ verified independently by CP-SAT, 252s, and CDCL with a unary-ladder encoding, 15s); $L(5) = 3$ (lower: CDCL, 2304s; upper: a window-3 witness at $N = 1024$, 995s). Moreover the minimal infeasible cap-coalition at $N = 256$ is exactly $\{15, 16\}$ (deletion-minimal): every scheme has $\delta(15) \geq 2$ or $\delta(16) \geq 2$ — the crowns of the two competing class towers ($15 = 1111_2$, apex of the $\equiv -1$ skeleton; $16 = 10000_2$, the block top, apex of the $\equiv 0$ skeleton).*

Theorem 28 (Divergence criterion). *If $L(m) \to \infty$ then S_{A} is not 3-permutable. Proof: an infinite valid permutation induces (Lemma 20) a normalized valid scheme, and by restriction a normalized valid scheme at every horizon 4^m (completions of values $\leq 4^m$ that lie beyond the horizon fall in the odd block $(4^m, 2 \cdot 4^m]$, out of team), and the ten displacements $\delta(3), \delta(4), \ldots, \delta(16)$ are constants of the infinite scheme. For m with $L(m) > \max_{v \leq 16} \delta(v)$ this is a contradiction, by the exactness of Theorem 19.* □

The conjectured growth $L(m) = m - 2$ matches the window-death schedule (window w dying at horizon 4^{w+3}) and the observed per-block descent of displacement in minimal solutions (δ profile $(m-2, m-3, \ldots, 1, 0)$ across blocks $4, 6, \ldots, 2m$). The missing step is a self-propagating squeeze $L(m+1) \geq L(m) + 1$; the $\times 4$ restriction alone yields only the diagonal transfer $L_{b+1}(m+1) \geq L_b(m)$ for the capped set $[1, 4^b]$.

13 The order gadget and the reduction

The machinery of Sections 10–12 compresses, for the canonical candidate S_{A}, into a one-scale-at-a-time statement about linear orders of single dyadic blocks. This section states that reduction precisely and proves it unconditionally — directly on a hypothetical permutation, without using the chunk calculus — and reports the machine verification of its finite instances; Section 14 then proves the required infeasibility by hand on the residue class $M \equiv 0 \pmod 8$ — which contains every dyadic scale — completing the proof of the main theorem (Theorem 35). Throughout, $b_j = M + j$ (*bottoms*) and $t_i = 2M - i$ (*tops*) denote elements of the interval $(M, 2M]$.

Definition 29 (Order gadget)**.** For an integer $M \geq 16$, the *order gadget* $\mathrm{OG}(M)$ asks for a linear order $\prec$ of the interval $(M, 2M]$ such that

- (i) *(no monotone AP)* for every arithmetic progression $a < b < c$ inside $(M, 2M]$, neither $a \prec b \prec c$ nor $c \prec b \prec a$ holds; and
- (ii) *(guards precede bottoms)* for $x \in \{15, 16\}$ and $1 \leq j \leq x/2$, the *guard* $t_{x-2j} = 2M+2j-x$ precedes the *bottom* $b_j = M + j$.

$\mathrm{OG}(M)$ is *infeasible* if no such order exists. (For $M \leq 15$ some guard coincides with the bottom it protects — $t_{x-2j} = b_j$ exactly when $M = x - j$, e.g. $t_{14} = b_1$ at $M = 15$ — and the gadget degenerates; we never use those scales. For $16 \leq M \leq 22$ a guard can coincide with a *different* bottom, which is harmless: every attack pair is a genuine pair for all $M \geq 16$.)

Three pieces of boundary arithmetic, each a one-line computation (machine-checked exhaustively in the repository, script `e96_reduction_check.py`), make the gadget exactly the trace of the infinite problem on one block: for $x \in \{15, 16\}$ the completion $2b_j - x$ lies in $(M, 2M]$ precisely for $j \leq x/2$, and then equals the guard t_{x-2j} (at $(x, j) = (16, 8)$ it is $2M$ itself, still inside the half-open block) — so (ii) lists *all* in-block completions of pairs (x, b_j) and nothing else; all other completions against 15 and 16 escape into the odd block above; and the guard–bottom pair's own downward completion $2b_j - t_{x-2j} = x$ is the attacking value itself, below the block, so the attack edges generate no unmodeled in-block constraint.

Theorem 30 (Order-gadget reduction; unconditional)**.** *If* $\mathrm{OG}(2^{2t-1})$ *is infeasible for infinitely many* $t \geq 4$*, then* S_{A} *is not* 3*-permutable.*

Proof. Suppose π is a 3-AP-free permutation of S_{A} and let $P = \max(\mathrm{pos}(15), \mathrm{pos}(16))$; note $15, 16 \in B_4 \subseteq S_{\mathrm{A}}$. Fix $t \geq 4$ and put $M = 2^{2t-1}$, so that $(M, 2M] = B_{2t} \subseteq S_{\mathrm{A}}$ and $15, 16 < M$.

We claim that some bottom b_j with $1 \leq j \leq 8$ has $\mathrm{pos}(b_j) \leq P$. Suppose not, and order the block by position: $u \prec v :\Longleftrightarrow \mathrm{pos}(u) < \mathrm{pos}(v)$. Constraint (i) is inherited from π. For (ii), take $x \in \{15, 16\}$ and $1 \leq j \leq x/2$. The pair (x, b_j) is placed increasingly ($\mathrm{pos}(x) \leq P < \mathrm{pos}(b_j)$, $x < M < b_j$), and its completion $z = 2b_j - x = t_{x-2j}$ lies in $(M, 2M] \subseteq S_{\mathrm{A}}$ and differs from b_j (else $M = x - j \leq 15$). If $\mathrm{pos}(z) > \mathrm{pos}(b_j)$, then (x, b_j, z) is a monotone 3-AP of π; hence $\mathrm{pos}(z) < \mathrm{pos}(b_j)$, i.e. $t_{x-2j} \prec b_j$. Thus $\prec$ witnesses $\mathrm{OG}(M)$, contradicting infeasibility.

So for every $t \geq 4$ with $\mathrm{OG}(2^{2t-1})$ infeasible, some element of $\{M + 1, \ldots, M + 8\}$, $M = 2^{2t-1}$, occupies a position $\leq P$. These blocks are disjoint, so infinitely many such t would place infinitely many distinct values among the first P positions — impossible. $\square$

Remark 31 (Chunk form; the crowns cannot defend every block)**.** In the coordinates of Theorem 19 the identical argument reads: in any valid scheme s, for every $t \geq 4$ with $\mathrm{OG}(2^{2t-1})$ infeasible, some bottom value in $\{M+1,\dots,M+8\}$ of block B_{2t} has stage $\leq \max(s(15), s(16))$. For if all eight had strictly larger stage, the induced order on the block (by stage, then fiber position) would satisfy $\mathrm{OG}(M)$: condition (i) holds for in-block triples directly by (A) and (B), and each attack $t_{x-2j} \prec b_j$ is forced whenever $s(x) < s(b_j)$ — by (A) the guard's stage cannot strictly exceed the bottom's, and at equal stages the forced-pair row $s(x) < s(y) = s(z) \Rightarrow z \prec y$ of (B) applies. The overflow step is then the finiteness of the fibers at stages $\leq \max(s(15), s(16))$; *no* displacement normalization (Lemma 20) is needed anywhere in this route. This is the precise sense in which the two *crowns* 15 and 16 of Section 12 cannot defend every dyadic block: they would have to be placed below the bottom-eight of all but finitely many blocks. We use only the direction scheme $\Rightarrow$ order; the converse (rebuilding a capped scheme from an OG witness, which would make the crown-cap infeasibility at a horizon *equivalent* to OG infeasibility at its top scale) has been verified only at the single scale $M = 128$, where the relevant family-MUS is single-block, and is not used here.

Proposition 32 (Machine verification; Cadical195)**.** $\mathrm{OG}(M)$ *is infeasible for every M with* $16 \leq M \leq 200$*, and for* $M = 512$*. In particular both tested scales of the dyadic family* $\{2^{2t-1}\}_{t\geq 4} = \{128, 512, 2048, \dots\}$ *are infeasible. The encodings assert (i) over all in-block triples and (ii) as unit clauses; order axioms are handled by lazy transitivity refinement, which is sound for infeasibility verdicts and was cross-validated against full eager $O(n^3)$ transitivity encodings at $M = 40, 44$ and — as an end-to-end recheck — at $M = 128$ (fresh eager instance,* 690,831 *clauses, UNSAT in about a second). The run at $M = 2048$ is unresolved (no verdict after* 200+ *refinement rounds), and the support growth reported below makes it unlikely that any finite sweep can settle the general statement.*

13.1 Attack cores: locating the load-bearing axioms

The hypothesis of Theorem 30 demands OG-infeasibility at infinitely many dyadic scales. The following machine facts locate a three-axiom core of the gadget that is already infeasible on exactly the residue class containing the dyadic family; Section 14 proves that core infeasibility by hand.

Proposition 33 (Attack cores; machine-checked)**.** *For every swept M (all of* $40 \leq M \leq 100$*, both parities, and* $M \in \{104, 108, 112, 120, 128, 150, 200\}$*), constraint (i) together with only the eleven attack units* $15\{1..7\} \cup 16\{1..4\}$ *— that is, $t_{15-2j} \prec b_j$ for $j \leq 7$ and $t_{16-2j} \prec b_j$ for $j \leq 4$ — is already infeasible. On residue subclasses, sharper cores hold at every swept scale: the four units $\{t_{13} \prec b_1,\ t_{11} \prec b_2,\ t_5 \prec b_5,\ t_{10} \prec b_3\}$ are infeasible with (i) exactly when* $M \equiv 0 \pmod 4$*, and the three units*

$$\mathrm{C3} \;=\; \{\, t_5 \prec b_5, \quad t_3 \prec b_6, \quad t_{10} \prec b_3 \,\}$$

are infeasible with (i) exactly when $M \equiv 0 \pmod 8$ *(satisfiable at every other swept residue, including* $M \equiv 4 \bmod 8$*).*

Every scale of the dyadic family is $\equiv 0 \pmod 8$, so Theorem 30 needs only the infeasibility of (i) $\wedge$ C3 on the class $M \equiv 0 \pmod 8$. That is exactly Theorem 36 below.

Remark 34 (Why a per-residue proof was to be expected)**.** The repository's parametric analysis (notes 27, 28, 30 and the `e87`–`e113` experiment suite) had established three structural facts that shaped the proof in Section 14. (1) Deletion-minimal certificate supports grow linearly with M (≈ 8 triples per unit of M), so no fixed finite list of parametric AP-triples certifies infeasibility at all scales; a uniform proof must use $\Theta(M)$-length mechanisms with $O(1)$ description — the ladder floods below are exactly that. (2) The forced-lemma layer behind small-scale refutations is parity-locked (reverse-forced at $M \not\equiv 0 \bmod 4$), so any uniform proof must stay within a residue class; the dyadic family lives in $M \equiv 0 \pmod 8$. (3) Fixed-denominator band-limit windows of the constraint system are satisfiable at all scales, so the obstruction is genuinely asymptotic and no compactness/limit argument over order types can carry it; the proof must be a scale-uniform schema, verified rung by rung.

14 The C3 core theorem and the main theorem

Theorem 35 (Main theorem)**.** *$S_{\mathrm{A}} = \bigcup_{k \text{ even}}(2^{k-1}, 2^k]$ is not 3-permutable: it admits no permutation (of order type ω) free of monotone 3-term arithmetic progressions. In particular the canonical dyadic partition $\mathbb{Z}^+ = S_{\mathrm{A}} \sqcup S_B$ is not a solution of the Erdős–Graham two-set problem: any partition witnessing a YES answer to problem #197, if one exists, must differ from the dyadic one.*

Proof. Every scale of the dyadic family satisfies $M = 2^{2t-1} \equiv 0 \pmod 8$ and $M \geq 128$ for $t \geq 4$. By Theorem 36 below, the C3 core — hence a fortiori the full gadget $\mathrm{OG}(M)$, whose attack list (Definition 29(ii)) contains the three axioms of C3 as the instances $(x, j) = (15, 5), (15, 6), (16, 3)$, and whose constraint (i) is AP-freeness — is infeasible at every such scale. Theorem 30 applies. □

The remainder of this section proves the core statement:

Theorem 36 (C3 core)**.** *For every $M \equiv 0 \pmod 8$ with $M \geq 16$, no AP-free linear order of the interval $(M, 2M]$ satisfies the three axioms*

$$\mathrm{C3} = \{\, A_1 \colon t_5 \prec b_5, \quad A_2 \colon t_3 \prec b_6, \quad A_3 \colon t_{10} \prec b_3 \,\}.$$

Here *AP-free* means condition (i) of Definition 29: no arithmetic progression $a < b < c$ inside the block occurs in monotone position order. The proof is a small toolkit of ladder lemmas (§14.1) followed by two forcing theorems (§14.2–§14.3). Every lemma application in the proofs has been machine-verified step by step at every $M \equiv 0 \pmod 4$ (resp. $\equiv 0 \pmod 8$) in $[12, 400]$ plus $M = 512, 1024$, and independently cross-validated; see Remark 43.

14.1 The ladder toolkit

AP-freeness is the *midpoint-extremal rule*: on every in-block AP (a, b, c), $c = 2b - a$, the midpoint b either precedes both endpoints or follows both. We use it as four unit rules (plus transitivity throughout):

R1: $a \prec b \Rightarrow c \prec b$, R3: $c \prec b \Rightarrow a \prec b$, R2: $b \prec c \Rightarrow b \prec a$, R4: $b \prec a \Rightarrow b \prec c$.

Write $m_0 = 3M/2$ (the arithmetic midpoint of the block: $b_j + t_j = 3M = 2m_0$ for all j, so every pair (b_j, t_j) mirrors through m_0). A *d-ladder* is a maximal run $w_0, w_1, \ldots$ of values of $(M, 2M]$ in arithmetic progression with difference d; the $d = 2$ ladders are the two parity classes, and the $d = 4$ ladders on odd values are *class A* $= \{v \equiv M + 1 \bmod 4\}$ and *class B* $= \{v \equiv M + 3 \bmod 4\}$.

Lemma 37 (Zigzag)**.** *Let $w_0, \ldots, w_r$ be consecutive rungs of a d-ladder and suppose $w_e \prec w_{e'}$ for some $|e - e'| = 1$. Then every rung w_i with $i \equiv e \pmod 2$ leads both its neighbors ($w_i \prec w_{i\pm1}$).*

Proof. Any three consecutive rungs form an AP with the middle rung as midpoint, so each rung either leads or trails both neighbors. The seed makes w_e a leader. If w_i leads, then from $w_i \prec w_{i+1}$, R1 on (w_i, w_{i+1}, w_{i+2}) gives $w_{i+2} \prec w_{i+1}$, and R4 on $(w_{i+1}, w_{i+2}, w_{i+3})$ gives $w_{i+2} \prec w_{i+3}$: w_{i+2} leads. Downward is the mirror argument. ☐

Lemma 38 (Phase dichotomy)**.** *In any linear order, every d-ladder is globally in one of its two zigzag phases: either all even-index rungs lead both neighbors, or all odd-index rungs do. Consequently the leader set of the ladder is one of its two half-classes modulo $2d$, adjacent rungs strictly alternate leader/trailer, and a proof may case-split on the phase and detach any conclusion derived in both branches.*

Proof. The order orients the adjacent pair (w_0, w_1) one way or the other; apply Lemma 37 to that seed. ☐

Lemma 39 (Transfer lock)**.** *Let M be even, $M \geq 12$ (below 12 the four rungs named here collide, e.g. $t_5 = b_3$ at $M = 8$). On the odd $d = 2$ ladder $w_i = M + 1 + 2i$ ($0 \leq i \leq M/2 - 1$) the four odd C3 values sit at $b_3 = w_1$, $b_5 = w_2$, $t_5 = w_{M/2-3}$, $t_3 = w_{M/2-2}$, and Lemma 37 locks the pair orientations together:*

$$M \equiv 0 \pmod 4 : \quad b_5 \prec b_3 \iff t_3 \prec t_5; \qquad M \equiv 2 \pmod 4 : \quad b_5 \prec b_3 \iff t_5 \prec t_3.$$

Proof. Each orientation of an adjacent pair seeds Lemma 37; read off whether the other pair's left member is a leader. At $M \equiv 0 \pmod 4$: $b_5 \prec b_3$ (i.e. $w_2 \prec w_1$) makes even indices leaders, and $M/2 - 2$ is even, so $t_3 = w_{M/2-2}$ leads its neighbor t_5. Conversely $t_3 \prec t_5$ seeds even-index leaders and $w_2 = b_5$ leads $w_1 = b_3$. The reverse orientations force the reverse conclusions, giving the biconditional; at $M \equiv 2 \pmod 4$ the parity of $M/2$ shifts the lock by one rung. ☐

Lemma 40 (Flood)**.** *Fix $g \in \{2, 4\}$ and a g-class C of $(M, 2M]$ (a parity class for $g = 2$; one of the odd mod-4 classes for $g = 4$) whose $d = g$ ladder is in a definite zigzag phase with leader set L. Let $c \in (M, 2M]$ be a center with $c \equiv C + g/2 \pmod g$, so that the mirror pairs $(c - e,\ c,\ c + e)$ with $e \equiv g/2 \pmod g$ have both members in C and form APs with midpoint c; call e admissible when both $c \pm e \in (M, 2M]$. Suppose some admissible e_0 carries a seed relation between c and a member $v \in \{c \pm e_0\}$. Then:*

- *(outward) if $c \prec v$, then $c \prec w$ for every $w \in C$ with $2c - w \in (M, 2M]$;*
- *(inward) if $v \prec c$, then $w \prec c$ for every such w.*

Moreover the conclusion holds in both zigzag phases of the C-ladder (phase-blindness), so by Lemma 38 it may be detached after a two-branch case split whenever the seed is available in both branches.

Proof. First, at each admissible e exactly one of $c+e$, $c-e$ is a leader: they differ by $2e \equiv g \pmod{2g}$, so they lie in the two different half-classes mod $2g$ of C, and L is one of them — whichever phase holds. Second, in a zigzag, adjacent rungs alternate and each leader leads both its neighbors.

Seed. From the relation at v, the mirror rule on the AP $(c-e_0, c, c+e_0)$ (R2/R4 outward, R1/R3 inward) gives the same-direction relation at the other member: both members of pair e_0 are related to c in the flood direction.

Outward, $e \to e+g$: let $x \in \{c \pm e\}$ be the leader of pair e. Its outward ladder-neighbor x' (same side, $|x'-c| = e+g$) satisfies $x \prec x'$; with $c \prec x$ known, transitivity gives $c \prec x'$, and the mirror rule floods $2c - x'$.

Outward, $e \to e-g$: let $w \in \{c \pm (e-g)\}$ be the trailer of pair $e-g$. Its outward neighbor w' (distance e, same side) is a leader and leads it: $w' \prec w$; with $c \prec w'$ known, $c \prec w$, and the mirror rule floods the other member.

Inward, $e \to e+g$: let x be the leader of pair $e+g$; it leads its inward neighbor x_{in} (distance e, same side): $x \prec x_{\text{in}} \prec c$, and the mirror rule floods the other member. *Inward,* $e \to e-g$: let x be the leader of pair $e-g$; it leads its outward neighbor (distance e): $x \prec x_{\text{out}} \prec c$; mirror floods the other member.

The admissible e form an interval of the residue class $g/2 \bmod g$, so the two inductions from e_0 reach every admissible e. Every step used only R1–R4 on genuine APs, transitivity, and zigzag edges of the given phase; the choice of side at each step is forced by the phase but *exists* in both phases. $\square$

Three instances of Lemma 40 are used below. **POLAR** ($g=2$, center m_0, class the odds; $M \equiv 0 \bmod 4$ makes m_0 even): seeded by the comparison of m_0 with t_5 (mirror b_5), it floods m_0 against *all* odd values (the mirror of b_l is t_l). The odd ladder's phase is pinned by the ambient hypotheses, so no dichotomy is needed. **P2-floods** ($g=2$, an odd center c near m_0, class the evens): seeded by c vs. m_0 at $e_0 = |c-m_0|$ (supplied by POLAR); the even ladder's phase is unknown — phase dichotomy, both branches. **G4-floods** ($g=4$, center $c \equiv \text{class}+2 \pmod 4$, class A or B): seeded at $e=2$ by the odd ladder's zigzag edges between c and $c \pm 2$; the $d=4$ ladder's phase is unknown — phase dichotomy, both branches.

The *mod-8 lock* of the whole problem sits in the center condition of the G4-floods: at $M \equiv 0 \pmod 8$ the two odd neighbors of m_0 satisfy $m_0 - 1 \equiv 3$ and $m_0 + 1 \equiv 1 \pmod 4$, so $m_0 - 1$ is a G4-center for class A and $m_0 + 1$ for class B. At $M \equiv 4 \pmod 8$ both congruences fail (and the odd-ladder leader statuses at $m_0 \pm 1$ invert), the flood seeds do not exist, and the proofs below evaporate — matching the machine fact that (i) $\wedge$ C3 is *satisfiable* there (Proposition 33).

14.2 Layer 1: two axioms force the odd-pair orientations

Theorem 41 (L1)**.** *Let $M \equiv 0 \pmod 4$, $M \geq 12$, and let $\prec$ be an AP-free order of $(M, 2M]$ satisfying $A_2\colon t_3 \prec b_6$ and $A_3\colon t_{10} \prec b_3$. Then $b_5 \prec b_3$ and $t_3 \prec t_5$.*

Proof. It suffices to refute $S\colon b_3 \prec b_5$; given $b_5 \prec b_3$, Lemma 39 supplies $t_3 \prec t_5$. Assume A_2, A_3, S. By Lemma 37 (seed S) the odd ladder's leaders are the offsets $\equiv 3 \pmod 4$. Define

$$c^* = \begin{cases} m_0+1, & M \equiv 0 \pmod 8 \\ m_0-1, & M \equiv 4 \pmod 8 \end{cases} \qquad c^{**} = \begin{cases} m_0-1, & M \equiv 0 \pmod 8 \\ m_0+1, & M \equiv 4 \pmod 8 \end{cases}$$

so that $c^* \equiv 1 \pmod 4$ is a G4-center for class B whose odd neighbors $c^* \pm 2$ ($\equiv 3$ mod 4) are odd-ladder leaders, and $c^{**} \equiv 3 \pmod 4$ is a G4-center for class A and is itself an odd-ladder leader. Split on the comparison (m_0, t_5).

Case I: $t_5 \prec m_0$. POLAR-inward: every odd value $\prec m_0$. Then: (1) $b_3 \prec c^*$ by the G4-inward flood at c^* over class B — seeds $c^* \pm 2 \prec c^*$ (the leaders $c^* \pm 2$ lead their trailing neighbor c^*); $b_3 \in$ class B with mirror $2c^* - b_3 \in \{t_1, t_5\}$ in the block; both $d{=}4$ phases. (2) $c^* \prec t_{10}$ by the P2-outward flood at c^* over the evens — seed $c^* \prec m_0$ (POLAR, $e_0 = 1$); t_{10} even with mirror $2c^* - t_{10} \in \{M{+}8, M{+}12\}$ in the block; both even phases. (3) A_3: $t_{10} \prec b_3$. Together: the 3-cycle $b_3 \prec c^* \prec t_{10} \prec b_3$, a contradiction.

Case II: $m_0 \prec t_5$. POLAR-outward: $m_0 \prec$ every odd value. Then: (1) $c^{**} \prec t_3$ by the G4-outward flood at c^{**} over class A — seeds $c^{**} \prec c^{**} \pm 2$ (c^{**} is an odd-ladder leader); $t_3 \in$ class A with mirror $2c^{**} - t_3 \in \{b_1, b_5\}$; both phases. (2) $b_6 \prec c^{**}$ by the P2-inward flood at c^{**} over the evens — seed $m_0 \prec c^{**}$ (POLAR, $e_0 = 1$); b_6 even with mirror $2c^{**} - b_6 \in \{2M{-}8, 2M{-}4\}$; both phases. (3) A_2: $t_3 \prec b_6$. Together: the 3-cycle $c^{**} \prec t_3 \prec b_6 \prec c^{**}$, a contradiction.

Both cases are contradictory, so S is impossible. □

Note that Case I consumes only A_3 and Case II only A_2; the boundary arithmetic (mirrors in the block, seed admissibility) holds down to $M = 12$.

14.3 The flip: the third axiom is anti-forced

Theorem 42 (FLIP)**.** *Let $M \equiv 0 \pmod 8$, $M \geq 16$, and let $\prec$ be an AP-free order of $(M, 2M]$ satisfying A_2, A_3 and $b_5 \prec b_3$. Then $t_5 \prec b_5$ is impossible: $b_5 \prec t_5$ ($= \neg A_1$) is forced.*

Proof. Assume $A_1\colon t_5 \prec b_5$ for contradiction. By Lemma 37 (seed $b_5 \prec b_3$) the odd ladder's leaders are the offsets $\equiv 1 \pmod 4$. Since $M \equiv 0 \pmod 8$: $m_0 - 1 \equiv 3 \pmod 4$ is a G4-center for class A whose odd neighbors $m_0 + 1$, $m_0 - 3$ (offsets $\equiv 1$ mod 4) are odd-ladder leaders; $m_0 + 1 \equiv 1 \pmod 4$ is a G4-center for class B and is itself an odd-ladder leader. Split on (m_0, t_5).

Case I: $t_5 \prec m_0$. POLAR-inward: every odd $\prec m_0$. Then: (1) $m_0 - 1 \prec t_{10}$ [P2-outward flood at $m_0 - 1$ over the evens; seed $m_0 - 1 \prec m_0$ (POLAR, $e_0 = 1$); mirror of t_{10} is $M + 8$; both phases]. (2) $m_0 - 1 \prec b_3$ [(1) $+A_3$]. (3) $m_0 - 1 \prec t_5$ [R4 on the AP (b_3, $m_0 - 1$, t_5): $b_3 + t_5 = 3M - 2 = 2(m_0 - 1)$]. (4) $m_0 - 1 \prec b_5$ [(3) $+A_1$]. (5) $b_5 \prec m_0 - 1$ [G4-inward flood at $m_0 - 1$ over class A; seeds $m_0 + 1 \prec m_0 - 1$ and $m_0 - 3 \prec m_0 - 1$ (leader edges); $b_5 \in$ class A at pair distance $M/2 - 6 \equiv 2 \pmod 4$ with mirror t_7 in the block; both phases]. (4) and (5) form a 2-cycle, a contradiction.

Case II: $m_0 \prec t_5$. POLAR-outward: $m_0 \prec$ every odd. Then: (1) $b_6 \prec m_0 + 1$ [P2-inward flood at $m_0 + 1$ over the evens; seed $m_0 \prec m_0 + 1$ (POLAR, $e_0 = 1$); mirror of b_6 is $2M - 4$; both phases]. (2) $t_3 \prec m_0 + 1$ [A_2 + (1)]. (3) $b_5 \prec m_0 + 1$ [R3 on the AP (b_5, $m_0 + 1$, t_3): $b_5 + t_3 = 3M + 2 = 2(m_0 + 1)$]. (4) $t_5 \prec m_0 + 1$ [A_1 + (3)]. (5) $m_0 + 1 \prec t_5$ [G4-outward flood at m_0+1 over class B; seeds $m_0+1 \prec m_0+3$ and $m_0+1 \prec m_0-1$ (m_0+1 is an odd-ladder leader); $t_5 \in$ class B at pair distance $M/2-6 \equiv 2 \pmod 4$ with mirror b_7; both phases]. (4) and (5) form a 2-cycle, a contradiction. □

Again Case I consumes A_3 and Case II consumes A_2, while A_1 is consumed symmetrically in both (step 4). The proof visibly fails at $M \equiv 4 \pmod 8$ in both cases: the centers $m_0 \mp 1$ land in the wrong mod-4 classes for their G4-floods and the odd-ladder leader statuses at $m_0 \pm 1$ invert, so neither seed set exists.

Proof of Theorem 36. $A_2 + A_3$ force $b_5 \prec b_3$ (Theorem 41; $M \equiv 0 \bmod 8 \subseteq 0 \bmod 4$). Then A_1 contradicts Theorem 42. □

Remark 43 (Machine verification of the schemas)**.** Although the proofs above are complete hand proofs, each is a *schema* whose instances at a given scale involve $\Theta(M)$ ladder steps; all have been machine-executed with strict per-step assertions (every AP's membership and arithmetic, every rule pattern, every leader/trailer and residue claim, every mirror bound, both branches of every phase dichotomy): Theorem 41 at every $M \equiv 0 \pmod 4$ in $[12, 400]$ plus $512, 1024$ (100 scales), Theorem 42 at every $M \equiv 0 \pmod 8$ in $[16, 400]$ plus $512, 1024$ (51 scales), and the $M \equiv 4 \pmod 8$ inapplicability at 35 scales (script `e113_c3_hand_proof.py`). An independent closure engine (plain R1–R4 + transitivity fixpoint, no knowledge of the schemas) refutes every branch of both case trees at 51 resp. 27 scales up to 512 (`e113b_closure_crossval.py`); and direct SAT checks of the two theorem *statements* at fresh scales never touched by the discovery loop ($M = 148, 212, 264, 328$ UNSAT; $M = 268, 332, \equiv 4 \bmod 8$, SAT) concur (`e114_theorem_spotcheck.py`). The earlier exhaustive base ledgers (AP + C3 UNSAT at every $M \equiv 0 \bmod 8$ in $[16, 256]$ and at 512) are now corroboration, not a dependency.

Remark 44 (What remains open)**.** The main theorem needs nothing beyond the class $M \equiv 0 \pmod 8$. The full order gadget is nevertheless conjectured infeasible at *every* $M \geq 16$ (machine-verified for $16 \leq M \leq 200$ and $M = 512$, Proposition 32); at residues $\not\equiv 0 \bmod 8$ the C3 core is satisfiable and other attack subsets take over (Proposition 33), so a proof would need per-residue analogues of Theorems 41–42 built from the same flood toolkit. This statement is not needed for Theorem 35. The Erdős–Graham problem #197 itself remains open: Theorem 35 eliminates the canonical candidate partition, but does not decide whether some other two-set partition works.

15 Historical note: suffix-stacked deferral

The deferral-state experiments (states exist at $M = 32, 64, 128$; class coherence; sink algebra) are subsumed by the chunk coordinates: prefix-nested semantics permit top-half deferral that rigid chunks forbid, which is why the law's witnesses evade Lemma 24 while no closed-form law survived depth 4.

Data availability

All experiment scripts, machine-audit checkers (including the mutation-test suite for the checkers themselves), solver logs, witness data, and DRAT certificates for the finite base cases are contained in the accompanying repository, `https://github.com/Wkasel/erdos197`, together with `reproduce.sh`, a single script that re-runs the complete verification stack from a pinned environment. The DRAT certificates cover the finite base instances only; the validity of the $\forall M$ schemas rests on the hand proofs of Sections 13–14 and their per-step machine audit (Remark 43).

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